TECHNICAL ASSIGNMENT

Build an ASP.NET MVC application that consumes NASA's APOD (Astronomy Picture of the Day) API, stores the returned data in a database, and displays the images in a gallery view.

Prerequisites

- Ensure the following before starting the assignment:
- Required Tools & Technologies
- .NET SDK
- Visual Studio VS Code
- SQL Server / SQL Express
- Basic understanding of:
    - ASP.NET MVC
    - DTOs
    - REST API
    - JSON serialization/deserialization

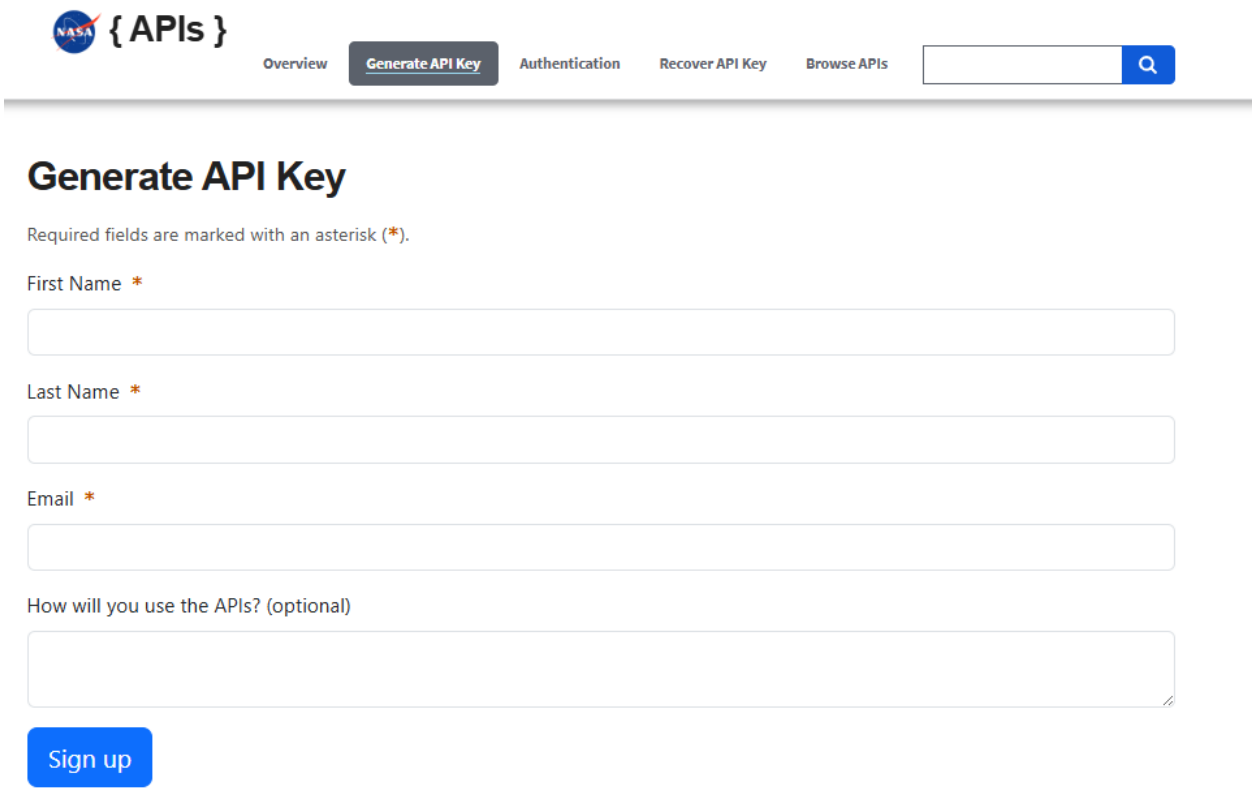
Assignment Requirements

1. Create a New ASP.NET MVC Project

- Set up a clean MVC project structure with Controllers, Models, Views, Services, etc.

2. Consume NASA APOD API

- Use the NASA APOD endpoint to fetch APOD data for a given date range.
    - NASA Open APIs Please do find the APOD API endpoint from here
- The response must be a JSON array which you will deserialize.


{ APIs }
Overview
Generate API Key
Authentication
Recover API Key
Browse APIs
Generate API Key
Required fields are marked with an asterisk (*).
First Name *
Last Name *
Email *
How will you use the APIs? (optional)
Sign up

Get API key

HTTP Request

`GET https://api.nasa.gov/planetary/apod`

concept_tags are now disabled in this service. Also, an optional return parameter *copyright* is returned if the image is not public domain.

Query Parameters

Parameter	Type	Default	Description
date	YYYY-MM-DD	*today*	The date of the APOD image to retrieve
start_date	YYYY-MM-DD	none	The start of a date range, when requesting date for a range of dates. Cannot be used with `date`.
end_date	YYYY-MM-DD	*today*	The end of the date range, when used with `start_date`.
count	int	none	If this is specified then `count` randomly chosen images will be returned. Cannot be used with `date` or `start_date` and `end_date`.
thumbs	bool	False	Return the URL of video thumbnail. If an APOD is not a video, this parameter is ignored.
api_key	string	DEMO_KEY	api.nasa.gov key for expanded usage

Example query

`https://api.nasa.gov/planetary/apod?api_key=DEMO_KEY`

Explore the query Prameters

3. Create DTO Classes

- Create DTOs matching NASA APOD API's JSON response properties.

4. Create SQL Database & Table

- Create a SQL Server database manually.
- Create a table to store APOD data.
- Include fields such as:
    - Date
    - Title
    - Explanation

    - Url
    - MediaType
    - ServiceVersion
    - SavedAt (timestamp)
    - *(Exact column names and types are up to your design.)*

5. Store API Response Using SQL Client (ADO.NET)

- Use SqlConnection, SqlCommand, SqlParameter, and raw SQL queries.
- Insert only new APOD entries (avoid duplicates).
- Use parameterized queries.

6. Retrieve Records Using SQL Client

- Use ADO.NET queries to read data from the APOD table.
- Convert SQL rows into objects that you will pass to the view.

7. Display Gallery on Home Page

- Display all saved APOD records in a gallery layout.
- Each block must have:
    - The image
    - The title

Expected Output

1. Git Repository

- Your submission must include:
- Source code
- Database script (SQL file to create the table)
- Instructions to run the project
- Instructions for setting NASA API key

2. Functional MVC Application

- Fetches APOD data
- Saves data using SqlClient
- Reads data using SqlClient
- Displays all images in a responsive gallery

Submission

Please provide:

- The GitHub repository link
- Database creation script
- Any additional notes or instructions